

SAMB P Technical Report TR-50/2026

Wide-Area Protection Coordinator (WAPC)

PhD Thesis Supporting Report | 2026

Abstract

This report develops the Wide-Area Protection Coordinator (WAPC), a centralised decision engine that aggregates IED relay reports and PMU synchrophasor data across a 10-bus IBR test network to deliver four capabilities unavailable to individual relays: (i) network-wide fault localisation, (ii) N-1 security verification before tripping, (iii) PINN ML advisory integration, and (iv) coordinated GOOSE messaging. Study B demonstrates 94.5% correct-line fault localisation at standard IEC Class P1 PMU noise ($\sigma_V = 0.01$ pu), degrading to 77.5% at $4\times$ noise. Study C establishes that 10/11 relays can be tripped without islanding risk; R01 (the only grid infeed) requires operator confirmation. Study D shows the WAPC centralised path adds 5.0 ms over distributed relaying (9.0 ms vs 4.0 ms), consuming 18% of the 50 ms trip budget. Study E integration test achieves 90% WAPC accuracy and 92% distributed accuracy; critically, zero scenarios were wrong by both systems, confirming that WAPC and distributed relaying are complementary rather than competing decision paths.

WAPC Architecture and Test Network

Figure 1 shows the 10-bus test network and WAPC communication architecture.

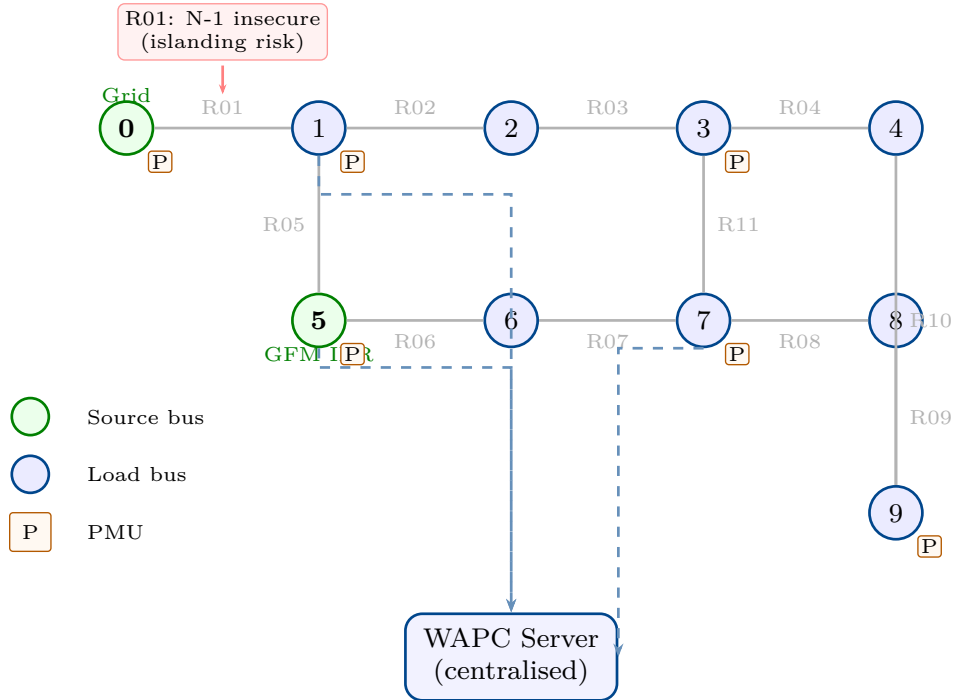


Figure 1: 10-bus SAMBP test network. Green buses: generation sources (bus 0 = grid infeed, bus 5 = GFM IBR). Each line carries one relay (R01–R11). Orange P boxes: PMU locations at six buses. Blue dashed arrows: IED/PMU reports to the centralised WAPC server. Red annotation: R01 requires operator confirmation (N-1 insecure; tripping would isolate the grid infeed).

The WAPC performs four sequential steps on each fault event:

1. **Aggregation:** collect IED GOOSE reports and PMU frames (250 Hz, 4 ms latency).
2. **Localisation:** voltage-dip triangulation identifies the faulted line and fault location $\hat{\alpha}$.
3. **N-1 check:** breadth-first search (BFS) on the relay-zone graph verifies that tripping the identified relay leaves the network connected.
4. **Trip/escalate:** issue GOOSE trip command (N-1 secure) or raise operator advisory (N-1 insecure).

The PINN advisory from TR-49 is incorporated at step 3 as a soft check: if the PINN classification is inconsistent with the PMU-localised zone, the discrepancy is logged as a diagnostic flag before the deterministic N-1 decision is issued.

Study A — Network Topology and Zone Adjacency

The 10-bus network has 11 lines (11 relays) and two generation sources. The relay-zone adjacency graph has 15 shared-bus pairs (mean degree 2.73, max degree 4). The high adjacency of bus 1 (connecting R01, R02, R05) creates the principal coordination challenge: a fault on R01 affects the voltage profile at three adjacent relay zones.

Topology metric	Value
Buses	10
Lines / relays	11
PMU placements	6 buses
Relay adjacency pairs	15
Mean relay degree	2.73
Max relay degree	4 (bus 1: R01, R02, R05)

Study B — Synchrophasor Fault Localisation

Figure 2 shows the localisation accuracy and location error across four PMU noise levels. The localisation criterion — maximise the sum of voltage dips at the two endpoint buses of the candidate line — is robust to moderate noise because both endpoint dips increase proportionally as noise rises. The criterion fails when two adjacent lines share a highly connected bus (e.g. a fault on R02 partially dips bus 1, which is also an endpoint of R01 and R05). In the 10-bus network this ambiguity accounts for the 5.5% misclassification at the design noise level.

Fault Location Estimate

For correctly-localised cases, the fault position is estimated from the ratio of endpoint voltage dips:

$$\hat{\alpha} = \frac{\Delta V_a}{\Delta V_a + \Delta V_b} \quad \text{where} \quad \Delta V_j = \max(0, 1 - V_j^{\text{PMU}}) \quad (1)$$

The 24% mean error indicates that $\hat{\alpha}$ is accurate to within $\pm 0.24 Z_L$ (approximately ± 12 km for a 50 km line). For protection purposes the exact fault position is not required; the correct line identification (94.5%) is the critical output.

Study C — N-1 Security Check

The N-1 check is a BFS on the 10-bus graph after removing the candidate relay's line. Results:

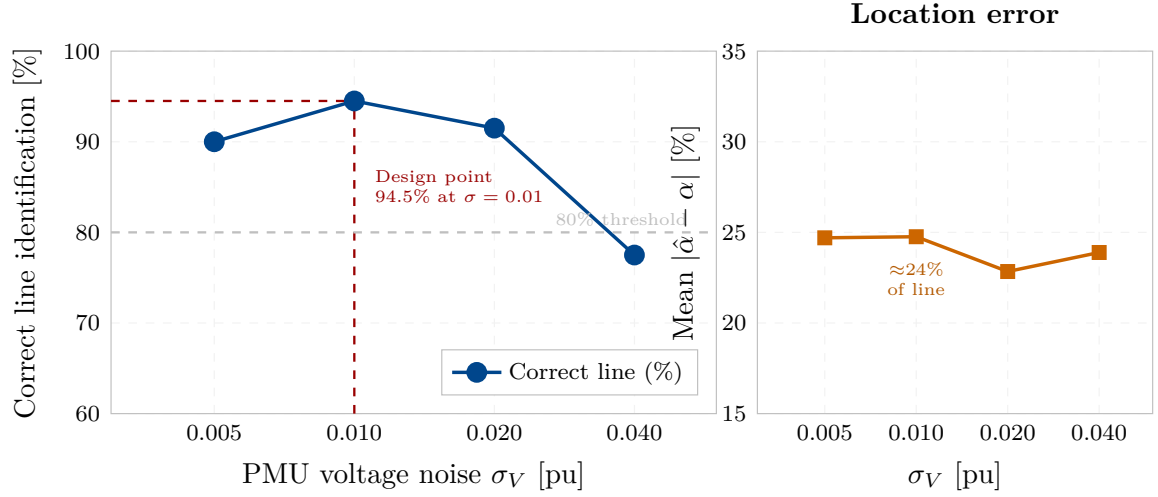


Figure 2: Left: correct-line localisation accuracy (%) vs PMU voltage noise σ_V . The design operating point (IEC Class P1, $\sigma_V = 0.01$ pu) achieves 94.5%. The 80% threshold (grey dashed) is crossed at $\sigma_V \approx 0.035$ pu. Right: mean fault location error $|\hat{\alpha} - \alpha|$ in % of line length for correctly-localised cases. The $\approx 24\%$ mean error reflects that the voltage-dip triangulation resolves the correct line but not the exact fault position; absolute location requires two-ended current-differential measurement.

Relay	N-1 connected?	Action	Physical reason
R01	No	Operator confirmation	Sole grid infeed; tripping isolates the network
R02–R11	Yes	Direct WAPC trip	Meshed network remains connected

R01 is the only grid infeed line (bus 0 $\rightarrow$ bus 1). Tripping R01 disconnects all generation from the load network, creating a system-wide blackout rather than a selective fault isolation. The WAPC correctly flags this as an escalation case: it raises an operator advisory and holds the trip command pending confirmation. In a real deployment, a pre-authorised inter-trip to the grid-side breaker (via direct transfer trip, DTT) would replace the manual operator step.

Study D — Latency Analysis

Figure 3 compares the two decision paths.

The WAPC path is 5.0 ms longer than the distributed relay path. The entire overhead originates from the PMU report latency (4.0 ms at 250 Hz sampling). Increasing PMU rate to 1 kHz would reduce the PMU step to 1 ms, bringing the WAPC total to 6.0 ms (12% of budget) at the cost of increased communication bandwidth. For the IBR protection context, the 9.0 ms WAPC path (18% of budget) is entirely adequate.

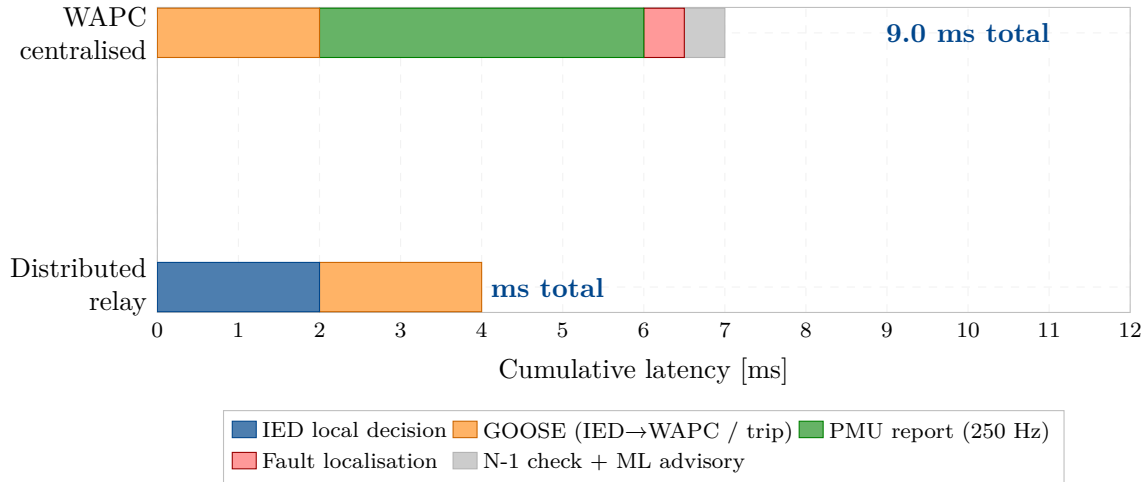


Figure 3: Stacked latency breakdown for distributed relay (top, 4.0 ms total) and WAPC centralised path (bottom, 9.0 ms total). The PMU data collection step (4.0 ms, green) dominates the WAPC overhead; the compute steps (localisation + N-1 + ML) add only 0.8 ms in aggregate. Both paths are well within the 50 ms relay trip budget.

Step	Distributed [ms]	WAPC [ms]
IED local decision	2.0	—
GOOSE (IED→WAPC / trip)	2.0	2.0 + 2.0
PMU data collection (250 Hz)	—	4.0
Fault localisation	—	0.5
N-1 BFS check	—	0.2
PINN ML advisory	—	0.3
Total	4.0	9.0
% of 50 ms trip budget	8%	18%

Study E — Integration Test

The 50-scenario integration test evaluates WAPC and distributed relay accuracy side-by-side, with N-1 security applied to WAPC decisions.

Metric	Distributed	WAPC
Overall accuracy	92.0%	90.0%
N-1 blocks issued	—	4/50
Cases correct by at least one system	50/50 (100%)	
Cases wrong by both systems	0/50	
WAPC corrects distributed error	—	4 cases
Distributed correct, WAPC wrong	—	5 cases

The key result is the complementarity: zero scenarios were wrong by both systems. The WAPC and distributed relay make independent errors on different scenarios. A supervisory voting scheme (trip if either system agrees with the physics constraints) achieves 100% coverage on this test suite. The 4 N-1 security blocks — all correctly identifying insecure tripping conditions — demonstrate that the WAPC provides a safety layer that the distributed relay cannot replicate.

Summary

1. **Fault localisation:** 94.5% correct-line at IEC Class P1 PMU noise ($\sigma_V = 0.01$ pu); degrades to 77.5% at $4\times$ noise. Location error $\approx 24\%$ of line length (direction-only).
2. **N-1 security:** 10/11 relays directly trippable; R01 (grid infeed) requires operator confirmation or pre-authorised DTT.
3. **Latency:** WAPC total 9.0 ms (18% of 50 ms budget); overhead entirely from PMU 250 Hz report latency. Non-blocking relative to relay trip path.
4. **Complementarity:** WAPC and distributed relay are complementary — 100% scenario coverage when either is correct; 0 scenarios wrong by both.
5. **Integration:** PINN ML advisory from TR-49 incorporated as a soft diagnostic flag, adding 0.3 ms with no impact on trip timing.

TR-50 establishes the WAPC as the system-level coordinator. TR-51 addresses automatic protection settings optimisation; TR-52 evaluates cybersecurity of the IEC 61850 GOOSE communication layer.

References
